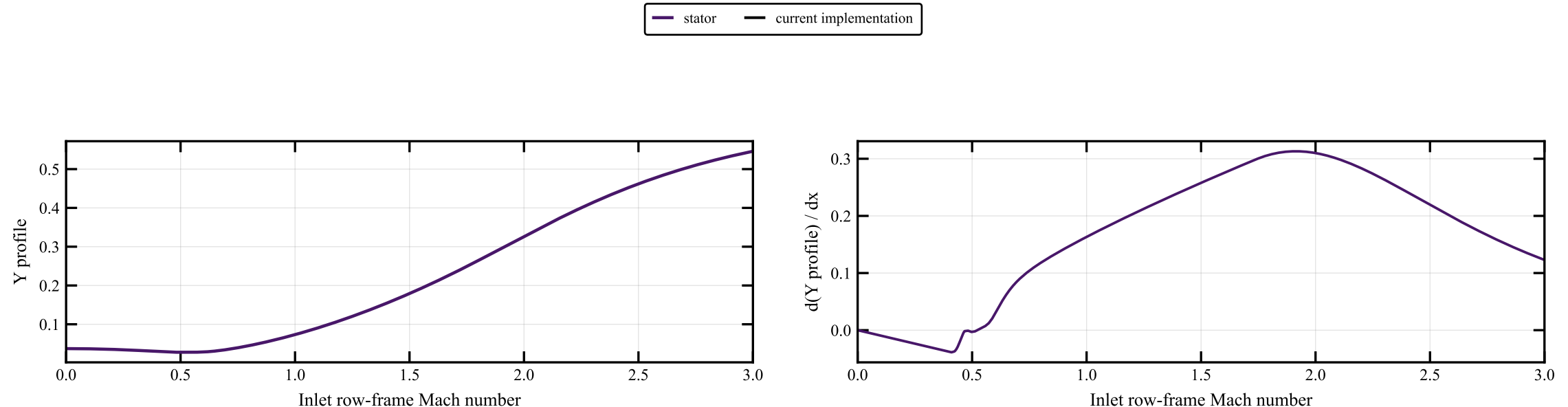ko profile Ma rel in — values and first derivatives